THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 10: **Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A ray strikes a plane mirror making an angle of 25 degrees with the mirror. The angle of reflection is: 1

(a) 155 degrees **(b)** 25 degrees **(c)** 50 degrees **(d)** 65 degrees

2. A concave mirror of focal length 10 cm has a radius of curvature: 1

(a) 40 cm **(b)** 20 cm **(c)** 5 cm **(d)** 10 cm

3. A convex lens converges a parallel beam to a point called the: 1

(a) optical centre **(b)** principal focus **(c)** pole **(d)** axis

4. The angle between the incident and the reflected ray is 80 degrees. The angle of incidence is: 1

(a) 10 degrees **(b)** 40 degrees **(c)** 160 degrees **(d)** 80 degrees

5. An object is placed at the centre of curvature of a concave mirror. The image is: 1

(a) virtual, inverted, smaller **(b)** real, erect, smaller

(c) virtual, erect, larger **(d)** real, inverted, same size

6. A concave lens forms an image of an object. The image is: 1

(a) real and inverted **(b)** virtual, erect and smaller

(c) real and larger **(d)** virtual and larger

7. A man walks towards a plane mirror at 2 m/s. The speed of his image towards him is: 1

(a) 4 m/s **(b)** 2 m/s **(c)** 0 **(d)** 1 m/s

8. An object is placed between the pole and the focus of a concave mirror. The image is: 1

(a) virtual, erect and enlarged **(b)** virtual and diminished

(c) real and diminished **(d)** real and inverted

9. An object is placed very far from a convex lens. The image forms: 1

(a) at the focus **(b)** at the lens

(c) far on the same side **(d)** at infinity

10. A girl stands 1.5 m from a plane mirror. If she walks 0.5 m towards it, the distance between her and the image is: 1

(a) 3 m **(b)** 1 m **(c)** 2 m **(d)** 0.5 m

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A ray of light makes an angle of 30 degrees with a plane mirror. What is the angle between the incident and reflected rays? 2

12. Why does the image formed by a convex mirror always appear smaller? 2

13. Why does a convex lens form a real image of a distant object? 2

14. Why is the mirror used by dentists concave? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ray falls on a plane mirror at 40 degrees to the normal. The mirror is turned by 10 degrees keeping the incident ray fixed. Find the new angle of incidence and the angle by which the reflected ray turns. 3

16. Why does the driver of a vehicle prefer convex rear-view mirrors? Mention any drawback. 3

17. Explain what happens to a parallel beam of light incident on a convex mirror and on a concave mirror. 3

OR

Explain why we see a well-polished wooden table's reflection but not that of a rough one. 3

18. Two plane mirrors are at 90 degrees. A ray strikes one mirror at 30 degrees to the normal. Trace the path and the final angle between the incident and the emergent ray. 3

OR

Why does the image formed on the retina appear inverted yet we see objects erect? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe an experiment to verify the laws of reflection. State the apparatus, procedure, observations and conclusion. 5

OR

A man 1.8 m tall stands in front of a plane mirror. Explain the minimum size of mirror and where to place it to see his full image. Show the geometry in words. 5

20. Two plane mirrors are inclined at 60 degrees with a candle placed between them. Find the number of images, and describe how they are formed. Compare with 90 degrees. 5

OR

Explain myopia and hypermetropia with the cause, correction and the lens used. Add a short note on the structure of the eye in relation to lens and retina. 5

END OF QUESTION PAPER — Total: 40 Marks